L E A R N Q U E S T

LINUX LABS DOCUMENTATION

MODULE 3

File, Directory & Link Management

Command-Line Practice Documentation

Linux Fundamentals Coursework

July 2026

1. Introduction

The Linux command line is the foundation of effective system administration, software development, and everyday computing on Linux-based systems. Unlike graphical interfaces, the terminal gives users direct, precise control over the file system — allowing files and directories to be created, organized, inspected, searched, compared, and linked with a small set of powerful commands.

This module documents hands-on practice with core Linux file-management utilities. Rather than presenting the commands as a single, undifferentiated list, they are organized below by functional category — Directory Management, File Management, Viewing Files, Searching, Comparing Files, and Links — so that related commands can be learned, referenced, and revised together. Each category includes a summary table of the commands practiced, followed by terminal screenshots demonstrating the corresponding output.

2. Learning Objectives

By the end of this module, the learner will be able to:

- Explain the role of the Linux command-line interface in managing files and directories.
- Create and remove directories and files using mkdir, touch, and rm.
- List and inspect file attributes using various options of the ls command.
- View file contents in different ways using cat, head, and tail.
- Search for text within files using grep, including case-insensitive matching.
- Compare files and directories line by line, by summary, or side by side using diff.
- Create, distinguish, and manage hard links and symbolic (soft) links using ln.

3. Learned Path

The commands in this module build on one another in a natural, practical sequence:

Step 1 — Set up the workspace: Create a directory to work in using mkdir, then populate it with files using touch.

Step 2 — Manage files: Practice removing files and directories with rm to keep the workspace clean.

Step 3 — View and inspect: Use the many options of ls, along with cat, head, and tail, to list files and read their contents.

Step 4 — Search: Use grep to locate specific text inside files quickly and accurately.

Step 5 — Compare: Use diff to identify differences between files and directories.

Step 6 — Link: Use ln to create hard and symbolic links, and observe how each behaves when the original file is modified or removed.

4. Directory Management

Command	Description
mkdir	Creates a new, empty directory at the specified location.

No dedicated terminal screenshot for mkdir was captured in this module; its effect is visible as the pre-existing "project" directory referenced in the screenshots below.

5. File Management

Command	Description
touch	Creates a new, empty file in the current directory (or updates its timestamp if it already exists).
rm	Deletes (removes) the specified file or directory from the file system.
rm -r	Recursively removes a directory and all of its contents.

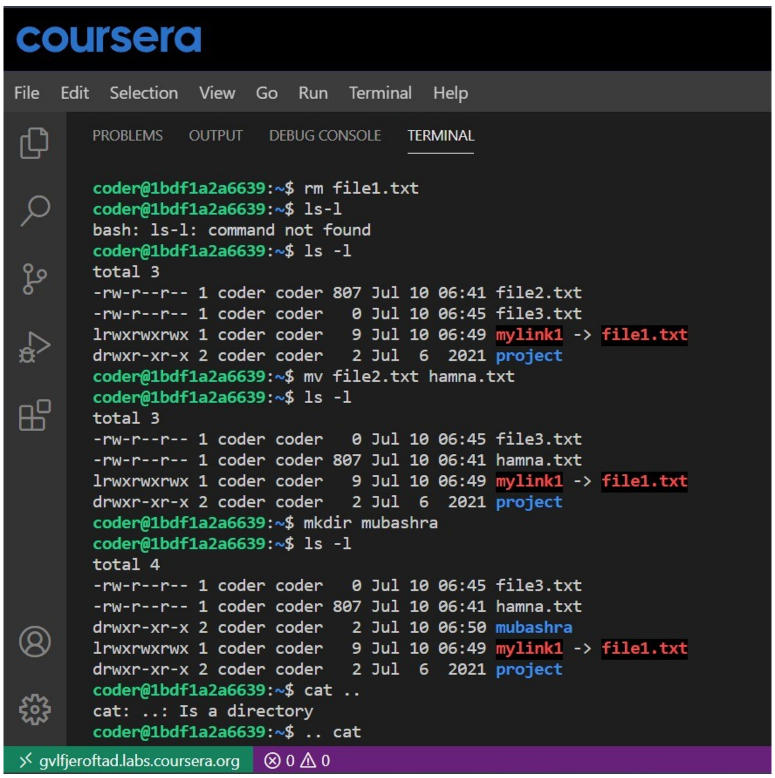

Figure 1 — Removing a file with rm and confirming with ls -l.

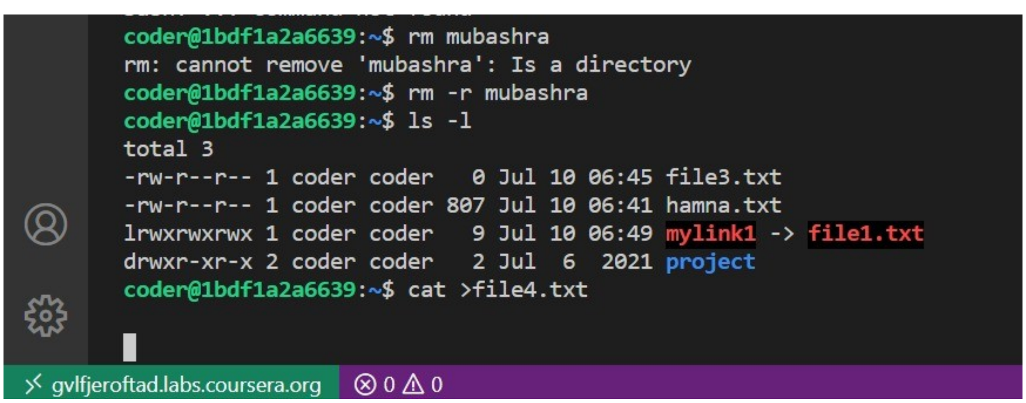

Figure 2 — Recursively removing a directory (rm -r) and creating a file with cat > file.

6. Viewing Files

Command	Description
ls -o	Lists files in long format without displaying the group name.
ls -g	Lists files in long format without displaying the owner name.
ls -i	Lists files along with their inode (index) numbers.
ls -s	Lists files together with their allocated size.
ls -lt	Lists files sorted by modification time, most recent first.
ls -lS	Lists files sorted by size, largest first.
ls -lr	Lists files in reverse order.
cat > file	Creates a new file and lets you type its contents directly.
cat f1 f2 > f3	Combines (concatenates) the contents of two files into a third file.
cat -n file	Displays file contents with line numbers.
cat -s file	Suppresses repeated blank lines in the output.
head -1 file	Displays the first N lines of a file (here, the first line).
tail -1 file	Displays the last N lines of a file (here, the last line).

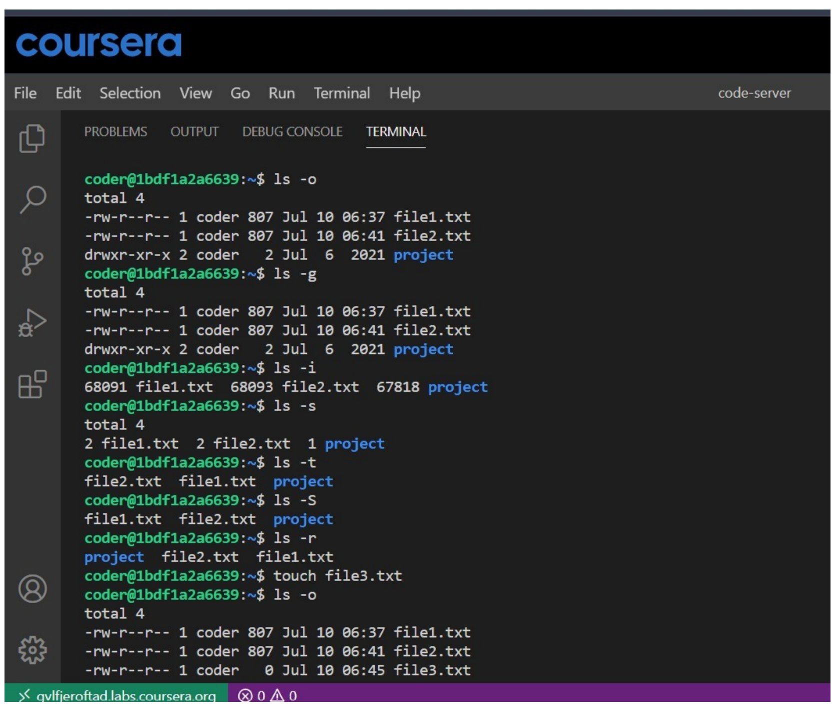

Figure 3 — ls -o listing files without the group column.

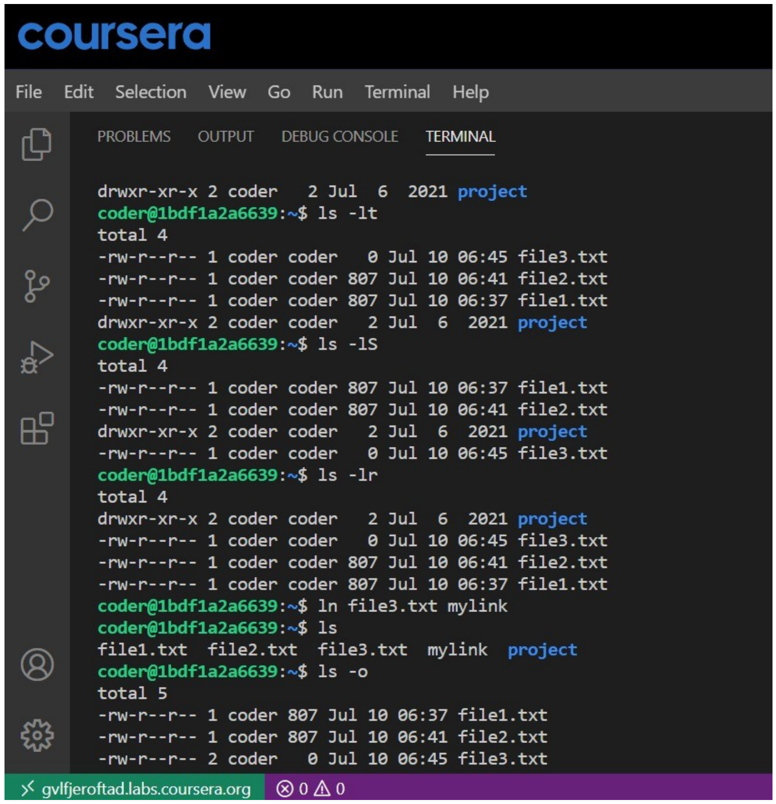

Figure 4 — ls -lt listing files sorted by modification time.

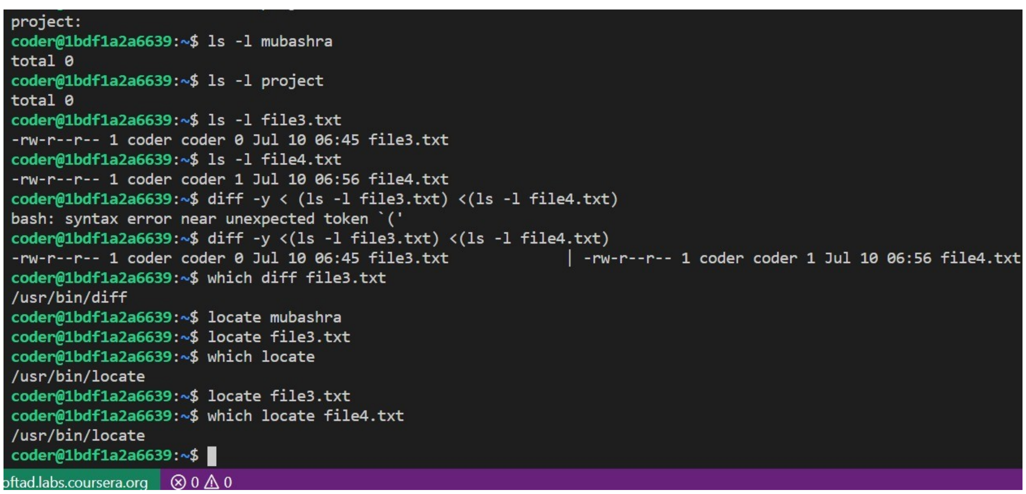

Figure 5 — ls -l used on individual files and directories.

7. Searching

Command	Description
grep -i "str" file	Searches for a string inside a file; -i makes the match case-insensitive.

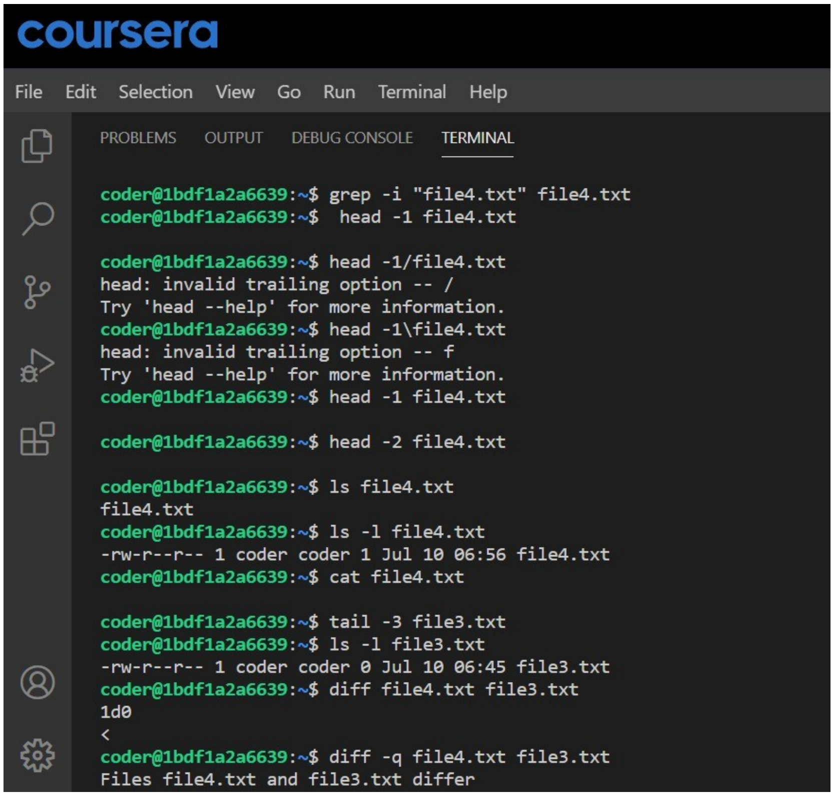

Figure 6 — grep -i searching a file for a string, alongside head -1.

8. Comparing Files

Command	Description
diff file1 file2	Compares two files line by line and reports the differences.
diff -q f1 f2	Reports only whether two files differ, without showing the details.
diff -r dir1 dir2	Recursively compares two directories, including their subdirectories and contents.
diff -y f1 f2	Displays the differences between two files side by side, in two columns.

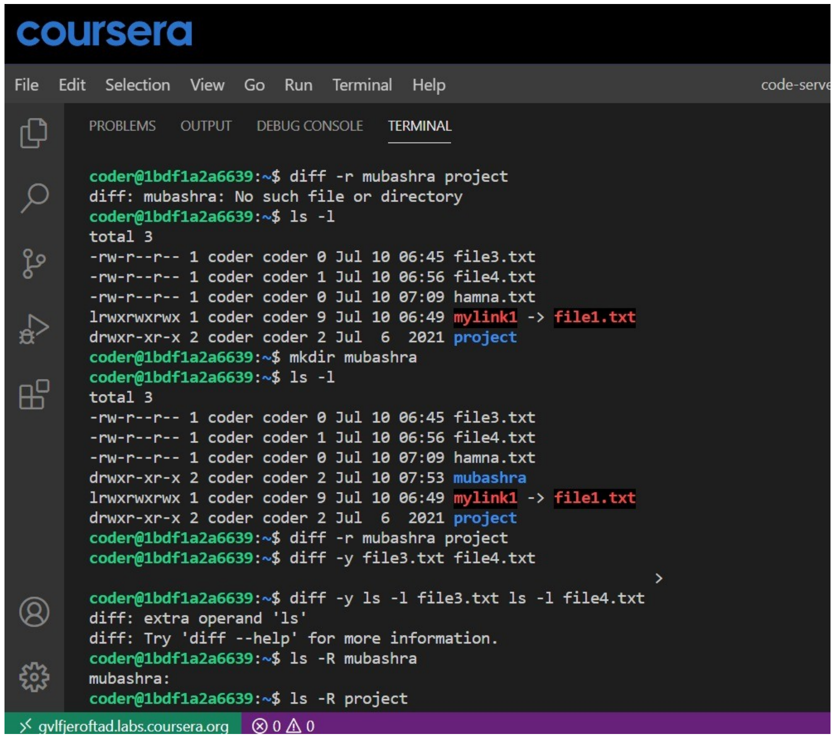

Figure 7 — diff -r comparing a missing directory against project.

9. Links

Command	Description
ln file link	Creates a hard link: a second name for a file that shares the same inode. Deleting one name does not remove the underlying data.
ln -s file link	Creates a symbolic (soft) link: a separate file that points to the target path. Deleting the original file breaks the link.

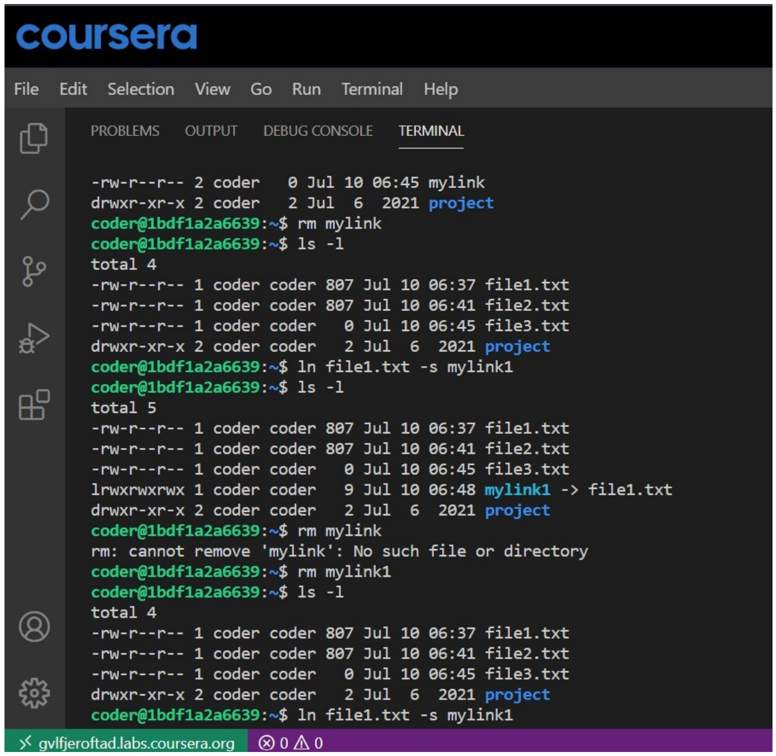

Figure 8 — Removing a link (mylink) and confirming the remaining files with ls -l.

10. Appendix — Additional Practice Screenshots

The screenshot below documents further exploration beyond this module's core command set (locate, which, and find), captured during the same practice session.

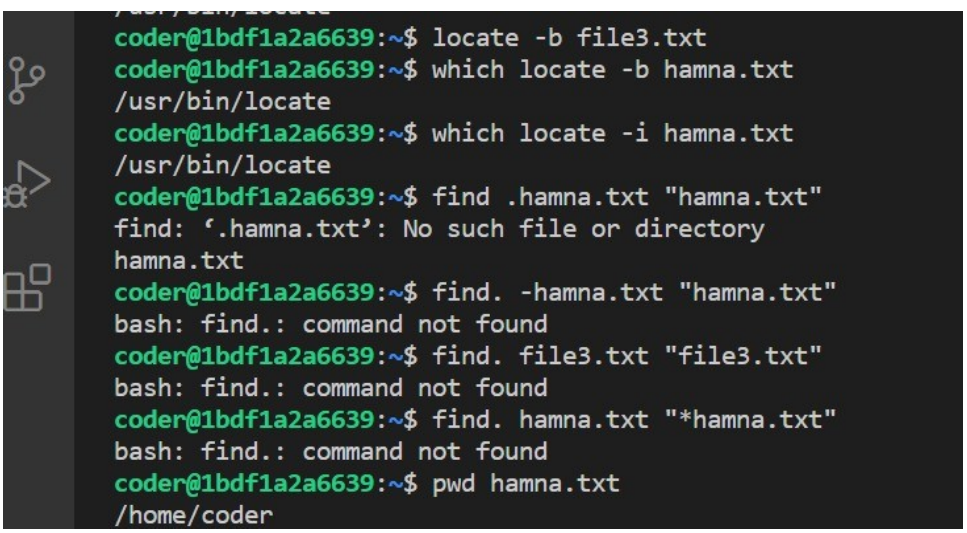

Figure 9 — Exploring locate, which, and find for file lookup.

11. Summary

Across this module, the practiced commands cover the full lifecycle of everyday file management on Linux: creating directories and files, removing them safely, listing and reading their contents in multiple formats, searching within them, comparing them, and linking them together. Together, these six categories form a practical foundation for working confidently in the Linux command-line environment.